BONGARD-LOGO: A New Benchmark for Human-Level Concept Learning and Reasoning

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Abstract

Humans have an inherent ability to learn novel concepts from only a few samples and generalize these concepts to different situations. Even though today's machine learning models excel with a plethora of training data on standard recognition tasks, a considerable gap exists between machine-level pattern recognition and human-level concept learning. To narrow this gap, the Bongard problems (BPs) were introduced as an inspirational challenge for visual cognition in intelligent systems. Despite new advances in representation learning and learning to learn, BPs remain a daunting challenge for modern AI. Inspired by the original one hundred BPs, we propose a new benchmark BONGARD-LOGO for human-level concept learning and reasoning. We develop a program-guided generation technique to produce a large set of human-interpretable visual cognition problems in action-oriented LOGO language. Our benchmark captures three core properties of human cognition: 1) context-dependent perception, in which the same object may have disparate interpretations given different contexts; 2) analogy-making perception, in which some meaningful concepts are traded off for other meaningful concepts; and 3) perception with a few samples but infinite vocabulary. In experiments, we show that the state-of-the-art deep learning methods perform substantially worse than human subjects, implying that they fail to capture core human cognition properties. Finally, we discuss research directions towards a general architecture for visual reasoning to tackle this benchmark.

1 Introduction

Human visual cognition, a key feature of human intelligence, reflects the ability to learn new concepts from a few examples and use the acquired concepts in diverse ways. In recent years, deep learning approaches have achieved tremendous success on standard visual recognition benchmarks [1, 2]. In contrast to human concept learning, data-driven approaches to machine perception have to be trained on massive datasets and their abilities to reuse acquired concepts in new situations are bounded by the training data [3]. For this reason, researchers in cognitive science and artificial intelligence (AI) have attempted to bridge the chasm between human-level visual cognition and machine-based pattern recognition. The hope is to develop the next generation of computing paradigms that capture innate abilities of humans in visual concept learning and reasoning, such as few-shot learning [4, 5], compositional reasoning [6, 7], and symbolic abstraction [8, 9].

The deficiency of standard pattern recognition techniques has been pinpointed by interdisciplinary scientists in the past several decades [3, 10]. Over fifty years ago, M. M. Bongard, a Russian computer

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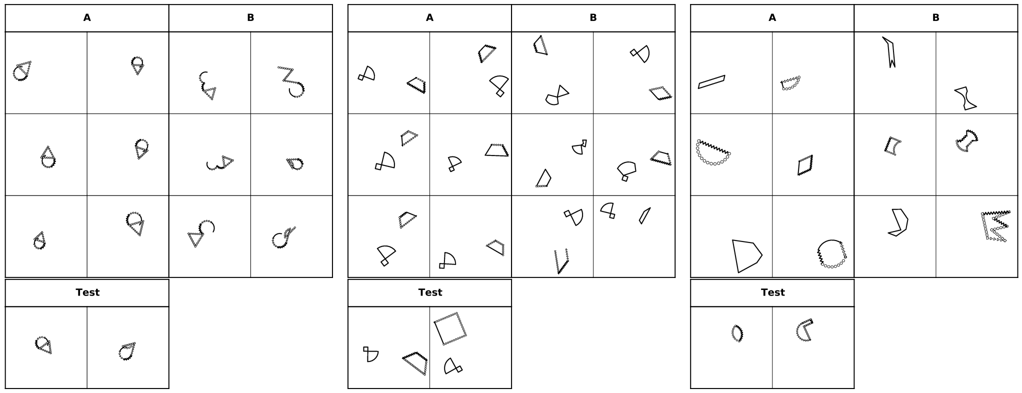

(a) free-form shape problem (b) basic shape problem (c) abstract shape problem

Figure 1: Three types of problems in BONGARD-LOGO, where (a) the free-form shape concept is a sequence of six action strokes forming an "ice cream cone"-like shape, (b) the basic shape concept is a combination of "fan"-like shape and "trapezoid", (c) the abstract shape concept is "convex". In each problem, set $\mathcal{A}$ contains six images that satisfy the concept, and set $\mathcal{B}$ contains six images that violate the concept. We also show two test images (left: positive, right: negative) to form a binary classification task. All the underlying concepts in our benchmark are solely based on shape strokes, categories and attributes, such as convexity, triangle, symmetry, etc. We do not distinguish concepts by the shape size, orientation and position, or relative distance of two shapes.

scientist, invented a collection of one hundred human-designed visual recognition tasks, now named the Bongard Problems (BPs) [11], to demonstrate the gap between high-level human cognition and computerized pattern recognition. Several attempts have been made in tackling the BPs with classic AI tools [12, 13], but to date we have yet to see a method capable of solving a substantial portion of the problem set. BPs demand a high-level of concept learning and reasoning that is goal-oriented, context-dependent, and analogical [14], challenging the objectivism (e.g., image classification by hyperplanes) embraced by traditional pattern recognition approaches [14, 15]. Therefore, they have been long known in the AI research community as an inspirational challenge for visual cognition. However, the original BPs are not amenable to today's data-driven visual recognition techniques, as this problem set is too small to train the state-of-the-art machine learning methods.

In this work, we introduce BONGARD-LOGO, a new benchmark for human-level visual concept learning and reasoning, directly inspired by the design principles behind the BPs. This new benchmark consists of 12,000 problem instances. The large scale of the benchmark makes it digestible by advanced machine learning methods in modern AI. To enable the scalable generation of Bongard-style problems, we develop a program-guided shape generation technique to produce human-interpretable visual pattern recognition problems in action-oriented LOGO language [16], where each visual pattern is generated by executing a sequence of program instructions. These problems are designed to capture the key properties of human cognition exhibited in original BPs, including 1) context-dependent perception, in which the same object may have fundamentally different interpretations given different contexts; 2) analogy-making perception, in which some meaningful concepts are traded off for other meaningful concepts; and 3) perception with a few samples but infinite vocabulary. These three properties together distinguish our benchmark from previous concept learning datasets that centered around standard recognition tasks [3, 7, 17, 18].

In our experiments, we formulate this task as a few-shot concept learning problem [4, 5] and tackle it with the state-of-the-art meta-learning [19–23] and abstract reasoning [17] algorithms. We find that all the models have significantly fallen short of human-level performances. This large performance gap between machine and human implies a failure of today's pattern recognition systems in capturing the core properties of human cognition. We perform both ablation studies and systematic analysis of the failure modes in different learning-based approaches. For example, the ability of learning to learn might be crucial for generalizing well to new concepts, as meta-learning methods largely outperform other approaches in our benchmark. Besides, each type of meta-learning methods has its preferred generalization tasks, according to their different performances on our test sets. Finally, we discuss potential research frontiers of building computational architectures for high-level visual cognition, driven by tackling the BONGARD-LOGO challenge.

2 BONGARD-LOGO Benchmark

Each puzzle in the original BP set is defined as follows: Given a set $\mathcal{A}$ of six images (positive examples) and another set $\mathcal{B}$ of six images (negative examples), the objective is to discover the rule (or concept) that the images in set $\mathcal{A}$ obey and images in set $\mathcal{B}$ violate. The solution to a BP is a logical rule stated in natural language that describes the visual concept presented in all images of set $\mathcal{A}$ but none of set $\mathcal{B}$. The original BPs consist of one hundred visual pattern recognition problems of black and white drawings. Through these carefully designed problems, M. M. Bongard aimed to demonstrate the key properties of human visual cognition capabilities and the challenges that machines have to overcome [11]. We highlight these key properties in detail in Section 2.2.

From a machine learning perspective, we can have multi-faceted interpretations of the BPs. Pioneer studies cast it as an inductive logic programming (ILP) problem [12] and a concept communication problem [13]. These two formulations typically require a significant amount of hand-engineering to define the logic rules or the language grammars, limiting their broad applicability. A more general and learning-oriented view of this problem is to cast it as a few-shot learning problem (as first mentioned in [24]), where the goal is to efficiently learn the concept from a handful of image examples. We are most interested in this formulation, as it requires the minimum amount of manual specification and likely leads to more generic approaches. In the next, we introduce our benchmark that inherits the properties of human cognition while being compatible with modern data-driven learning tools.

2.1 Benchmark Overview

We developed the BONGARD-LOGO benchmark that shares the same purposes as the original BPs for human-level visual concept learning and reasoning. Meanwhile, it contains a large quantity of 12,000 problems and transforms concept learning into a few-shot binary classification problem. Figure 1 shows some examples in BONGARD-LOGO. For example, in Figure 1c, set $\mathcal{A}$ contains six image patterns which are all convex shapes and, set $\mathcal{B}$ contains six image patterns which are all concave shapes. The task is to judge whether the pattern in the test image matches the concept (e.g., convex vs concave) induced by the set $\mathcal{A}$ or not. As the same concept might lead to vastly different patterns, a successful model must have the ability to identify the concept that distinguishes $\mathcal{A}$ and $\mathcal{B}$. The problems in BONGARD-LOGO belong to three types based on the concept categories:

Free-form shape problems In the first type of problems, we have 3,600 *free-form shape* concepts, where each shape is composed of randomly sampled action strokes. The rationale behind these problems is that the concept of an image pattern can be *uniquely* characterized by the action program that generates it [3, 25]. Here the latent concept corresponds to the sequence of action stokes, such as straight lines, zigzagged arcs, etc. Among all 3,600 problems, each has a *unique* concept of strokes, and the number of strokes in each shape varies from two to nine and each image may have one or two shapes. As shown in Figure 1a, all images in the positive set form a one-shape concept and share the same sequence of six strokes that none of images in the negative set possesses. Note that some negative images may have subtle differences in stokes from the positive set, such as perturbing a stroke from `straight_line` into `zigzagged_line`, and please see Appendix D.1 for more examples. To solve these problems, the model may have to implicitly induce the underlying programs from shape patterns and examine if test images match the induced programs.

Basic shape problems The other 4,000 problems in our benchmark are designed for the *basic shape* concepts, where the shapes are associated with 627 human-designed shape categories of large variation. The concept corresponds to recognizing one shape category or a composition of two shape categories presented in the shape patterns. Similar to the free-form shape problems, each problem has a unique basic shape concept. One important property of these problems is to test the analogy-making perception (see Section 2.2) where, for example, `zigzag` is an important feature in free-form shapes but a nuisance in basic shape problems. Figure 1b illustrate an instance of basic shape problems, where all six images in the positive set have the concept: a combination of "fan"-like shape and "trapezoid", while all six images in the negative set are other different combinations of two shapes. Note that positive images in Figure 1b may have zigzagged lines or lines formed by a set of circles (i.e., different stroke types), but they all share the same basic shape concept.

Abstract shape problems The remaining 4,400 problems are aimed for the *abstract shape* concepts. Each shape is sampled from the same set of human-designed 627 shape categories. But in contrast to the previous type, these concepts correspond to more abstract attributes and their combinations. The